

# Kabir Singh

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## ACADEMIC EDUCATION

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- Albert Dorman Honors College (ADHC)** and Jordan Hu College of Sciences and Liberal Arts (JHCSLA), New Jersey Institute of Technology, Newark, New Jersey, USA Sep '23 – May '27 (Expected)
- Bachelor of Science; Major – **Mathematical Biology**
  - Minors – **Biomedical Engineering in Technology, Philosophy, History**
  - ADHC Honors Tracks – **Medical Humanities, Global Studies**
  - *Academic Excellence, Honors Merit, Honors Residential Scholarship Recipient (Tuition + Board)*
- Faculty of Engineering, Lunds Tekniska Högskola, **Lunds Universitet**, Lund, Sweden Jan '25 – Jun '25
- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3)

## RESEARCH

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- Undergraduate Researcher, Mathematical Neuroscience. PI: Dr. Horacio Rotstein, NJIT Sep '26 – Present
- Deriving reduced mathematical descriptions of biophysical neuronal models to characterize their dynamic properties
- Undergraduate Researcher, **History of Medicine and Bioethics**. PI: Dr. Stephen Pemberton, NJIT Aug '26 – Present
- Synthesizing legal, clinical, and cultural developments in American end-of-life care from the late twentieth century to the present
  - Developing original scholarship for submission to a peer-reviewed humanities journal
- Undergraduate Researcher, Computational Bionanotechnology. PI: Dr. Stefano Angioletti-Uberti, Imperial College London Jun '26 – Present
- **BAGEL: Biomolecular Algorithm for Guidance in Energy Landscape**
  - Computational modelling of soft-matter and biomaterial systems
  - Applying statistical mechanics, molecular dynamics, and machine-learning methods to problems in enzyme design and broader protein design
- Undergraduate Researcher, **Computational Biology Team Lead** at KumarLab. PI: Dr. Vivek Kumar, University of Houston (Prev. NJIT) Oct '23 – Present
- Lead undergraduate and graduate students on designing and developing peptide-based therapeutics
  - Started at NJIT and continued computational work with lab relocation to Houston
  - Taught 15+ associates in lab
  - Received > \$15,000 in direct funding for projects
- Undergraduate Researcher, **Mathematical Modeling**. PI: Dr. James Maclaurin, NJIT Oct '25 – May '26
- Applying Hawkes self-exciting point process to model secondary nucleation dynamics in amyloid formation

## FUNDING

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- Undergraduate Research Opportunities Programme (UROP), Imperial College London Jun '26 – Jul '26
- Competitive research bursary awarded through the Department of Materials, Faculty of Engineering
  - Project: Using protein language models to mimic natural ligands. Supervisor: Prof. Stefano Angioletti-Uberti
  - Personal compensation included
- NSF I-Corps Regional Program, Northeast Hub May '26 – Jun '26
- Co-Founder & Technical Lead, Axioma Therapeutics
  - Selected into competitive NSF cohort; led customer discovery and lean-startup validation for cyclic peptide therapeutics targeting degenerative eye diseases
  - Project funding awarded to support customer-discovery activities

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| Grace Hopper AI Research Institute, Computational Protein Design                                                                                                                                                                                                                                                                                         | May '25 – Aug '25 |
| <ul style="list-style-type: none"> <li>– NJIT grant to develop a course on computational protein design</li> <li>– Teaching Assistant</li> </ul>                                                                                                                                                                                                         |                   |
| The Honors College Moonshot Grants for Engaged Scholarship                                                                                                                                                                                                                                                                                               | Jun '25 – Jul '25 |
| <ul style="list-style-type: none"> <li>– Endowed award to develop interdisciplinary projects that address important social and cultural questions</li> <li>– Designing Male Contraceptive</li> <li>– Award providing labs with funds to support undergraduate-led projects</li> <li>– Personal compensation included. One recipient per annum</li> </ul> |                   |
| Male Contraceptive Initiative (MCI)                                                                                                                                                                                                                                                                                                                      | Jan '25 – Jun '25 |
| <ul style="list-style-type: none"> <li>– Award by the Youth Advisory Board providing funding to support grantee-led project</li> <li>– Designing Male Contraceptive</li> <li>– Personal compensation included</li> </ul>                                                                                                                                 |                   |
| NJIT Undergraduate Research and Innovation (URI)                                                                                                                                                                                                                                                                                                         | Oct '24 – May '25 |
| <ul style="list-style-type: none"> <li>– Student Seed Grant award for a Phase-2 longitudinal project on application-based research funding lab to support undergraduate-led project</li> <li>– Developing cyclic peptide therapeutic for Multiple Sclerosis</li> </ul>                                                                                   |                   |
| Travel Award – National Contraception Meeting                                                                                                                                                                                                                                                                                                            | May '25           |
| <ul style="list-style-type: none"> <li>– Awarded by the Male Contraceptive Initiative for travel to Durham, NC, supporting a conference presentation</li> </ul>                                                                                                                                                                                          |                   |
| NJIT Honors Summer Research Institute (HSRI)                                                                                                                                                                                                                                                                                                             | Jun '24 – Jul '24 |
| <ul style="list-style-type: none"> <li>– Grant award by ADHC and URI providing funds to support undergraduate-led projects</li> <li>– A combination of Multiple Sclerosis project and Male Contraceptive project</li> <li>– Personal compensation included</li> </ul>                                                                                    |                   |

## PUBLICATIONS

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### Peer-Reviewed

- C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis. *European Journal of Medicinal Chemistry*, 119173 (2026). doi:10.1016/j.ejmech.2026.119173

### Manuscripts in Preparation

- **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. Machine learning for rational design of a \*\*\*\*\* targeting cyclic peptide-based male contraceptive. Publication [in preparation], **Patent application** [in preparation], 2026
- J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. Academic Publishing as a Launchpad: How Publishing Research in an Undergraduate Journal Shapes Future Careers. Publication [in preparation], 2026

## PRESENTATIONS

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| M. Kotta, G. Victorio, G. Karmacharya, <b>K. Singh</b> , N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor.                          |                   |             |
| UH (Undergraduate Research Day)                                                                                                                                                                                                     | Poster + Podium   | 2026        |
| <b>K. Singh</b> , C. Heffernan, J. Dodd-O, V. Kumar. Development of a novel male contraceptive using cyclic peptides to inhibit fertilization.                                                                                      |                   |             |
| AAP/ASCI/APSA Joint Meeting                                                                                                                                                                                                         | Poster            | 2026        |
| NJIT AI Exploration Day                                                                                                                                                                                                             | Poster            | 2026        |
| NJIT ADHC (Honors Interdisciplinary Research Forum) <b>First Place</b>                                                                                                                                                              | Poster            | Fall 2025   |
| NCM (National Contraception Meeting)                                                                                                                                                                                                | Poster            | 2025        |
| NJIT ADHC (Honors Interdisciplinary Research Forum)                                                                                                                                                                                 | Poster            | Spring 2025 |
| NJIT (Dana Knox Research Showcase)                                                                                                                                                                                                  | Poster            | 2025        |
| NJIT (Undergraduate Research and Innovation)                                                                                                                                                                                        | Abstract + Podium | 2025        |
| BMES (Biomedical Engineering Society) [Co-Presenter]                                                                                                                                                                                | Abstract + Poster | 2024        |
| NJIT (Undergraduate Research and Innovation)                                                                                                                                                                                        | Abstract + Podium | 2024        |
| <b>K. Singh</b> , J. Lau, H. Gerber, A. Alibilli, L. Goralsky. Mitigating Knowledge Drain: A Case Study in Optimizing Workflow and Training in a High-Turnover, International Peer-Reviewed Undergraduate Science Academic Journal. |                   |             |
| NCUR                                                                                                                                                                                                                                | Poster            | 2026        |

## ACADEMIC SERVICE

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| <b>22<sup>nd</sup> Editor-in-Chief</b> , Associate Editor (Former), Journal of Young Investigators                                                                                                                                                                          | Dec '24 – Present                       |
| <ul style="list-style-type: none"><li>– Manage timely publication of high-quality scientific content</li><li>– Develop initiatives to further the journal's mission</li><li>– Former specialization in Biology, Physical Sciences, and Social Science Departments</li></ul> |                                         |
| <b>Research Chair</b> , American Association of Physicians of Indian Origin (AAPI) PreMed Board                                                                                                                                                                             | Jun '25 – Present                       |
| <ul style="list-style-type: none"><li>– Lead committee and organize events to achieve research-related objectives on a national level to support students, researchers, physicians, and more of an Indian origin</li></ul>                                                  |                                         |
| <b>Board Member</b> , MCI Youth Advisory Board (YAB)                                                                                                                                                                                                                        | Aug '25 – Present                       |
| Member, Pre-Medical Ad Hoc Committee, APSA                                                                                                                                                                                                                                  | Mar '26 – Present                       |
| Member, Goldwater Ambassadors Program                                                                                                                                                                                                                                       | Aug '25 – Present                       |
| <ul style="list-style-type: none"><li>– Selected into competitive process as a mentee, mentored by Goldwater Scholars</li><li>– Program specially designed for students from underrepresented or underserved backgrounds</li></ul>                                          |                                         |
| Member, ARTISAN Program                                                                                                                                                                                                                                                     | Apr '25 – Present                       |
| <ul style="list-style-type: none"><li>– Inducted in the NSF ART Integrated Student Ambassador Network (ARTISAN) program, based off commitment to translational research in my work</li></ul>                                                                                |                                         |
| Member, Inclusive Excellence Student Advisory Board                                                                                                                                                                                                                         | Feb '25 – Present                       |
| Member, Multifaith Council                                                                                                                                                                                                                                                  | Jan '25 – Present                       |
| Member, Student Conduct Board                                                                                                                                                                                                                                               | Dec '24 – Present                       |
| <ul style="list-style-type: none"><li>– Serve as adjudicator on academic and behavioral cases in student conduct hearings, along with other members of the board, including faculty and NJIT administration</li></ul>                                                       |                                         |
| Chair, Chair-in-Training (Former), Highlander Integrity Council                                                                                                                                                                                                             | Sep '24 – Present                       |
| Organizer, 'It's Different for Girls' Documentary Screening and Panel, NJIT ADHC and CSLA Co-Sponsored Colloquium                                                                                                                                                           | Mar '26                                 |
| NJIT Representative, BioNJ 33 <sup>rd</sup> Annual Dinner Meeting & Innovation Celebration                                                                                                                                                                                  | Feb '26                                 |
| Grant Reviewer, MCI YAB Undergraduate Project Grant                                                                                                                                                                                                                         | Dec '25 – Jan '26                       |
| Teaching Assistant, Honors First Year Seminar                                                                                                                                                                                                                               | Sep '24 – Dec '24,<br>Sep '25 – Dec '25 |
| Invited Speaker, NJIT First Aid Squad                                                                                                                                                                                                                                       | Sep '25, Oct '25                        |
| <ul style="list-style-type: none"><li>– Presented to NJIT Board of Trustees as well as the President's Town Hall with squad leadership at the request of NJIT President Teik Lim</li></ul>                                                                                  |                                         |
| Steering Committee Lead, ADHC First Year Retreat                                                                                                                                                                                                                            | Mar '25 – Aug '25                       |
| Teaching Assistant, Grace Hopper AI Research Institute                                                                                                                                                                                                                      | May '25 – Aug '25                       |
| <ul style="list-style-type: none"><li>– Computational Protein Design</li></ul>                                                                                                                                                                                              |                                         |
| Featured Speaker, ADHC Advising Day                                                                                                                                                                                                                                         | Jun '24 & '25                           |
| Advisor, President (Former), Secretary (Former), Honors Hall Council                                                                                                                                                                                                        | Feb '24 – May '25                       |
| ADHC Representative, National Collegiate Honors Council Annual Conference                                                                                                                                                                                                   | Nov '24                                 |

## LECTURES & FEATURES

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- Lecture Series. Ethics in Science and Research, Science Mentorship Institute
- Session Moderator. Student Oral Presentation (Communications). NCUR, 2026
- Featured Presenter. 'Navigating peer review and the student journal landscape'. "Shaping Asia's Future: Development, Innovation, and Society in Transition" organized by Nazarbayev University, Asian Universities Alliance 2026
- Featured Panelist. 'It's Different for Girls' Documentary Screening and Panel. NJIT ADHC and CSLA Co-Sponsored Colloquium, NJIT 2026
- NJIT Makes: Undergraduate Research and Innovation, NJIT Promotional Material. NJIT, 2026
- Albert Dorman Honors College, Honors in Action 2025 (Research, Award, and Study Abroad Experience). ADHC Publication. NJIT, 2025

- Nayib Morán. ‘150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum’. Article Feature. NJIT News, 2025
- Albert Dorman Honors College, Honors in Action 2024 (Research and Study Abroad Experience). ADHC Publication. NJIT, 2024
- Gurnoor Kaur. ‘Undergraduate Student Researchers on Campus’. Article Feature. NJIT (Student Newspaper – The Vector), 2024
- K. Singh. ‘Introduction to PyMOL and Linear Peptide Design’. Lecture. NJIT (Genetically Engineered Machines), 2024

## CERTIFICATIONS & LICENSES

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- New Jersey EMT License
- ACS “Stop the Bleed” Certification & Instructor License
- OSHA General Industry Safety and Health Certification
- CITI Biomedical Research Certification
- CITI RCR Basic Certification
- American Heart Association BLS & CPR for Adult/Child/Infant Certification

## LEADERSHIP

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| Ethics Coordinator, Mentor (Neuro + Bio), Ethics Curriculum Dev (Former), Admission Officer, Symposium Committee Organizer, Grant Writer, Science Mentorship Institute (Sci-Mi) | May '24 – Present                       |
| Public Relations Chair, EMT, NJIT First Aid Squad                                                                                                                               | Oct '25 – Present,<br>Jan '24 – Present |
| Stop the Bleed Instructor (Hybrid English–Punjabi), NJ Gurdwaras, NJ Office of Homeland Security and Preparedness (NJOHSP) & Sikh Youth Alliance                                | Jan '26 – Present                       |
| Khalsa School Virsa (Sikh History) Teacher, Sikh Sabha of New Jersey                                                                                                            | Sep '20 – Present                       |
| Youth Camp Organizer and Coordinator, Sikh Sabha of New Jersey                                                                                                                  | Aug '21 – Present                       |
| Content Manager, The Mardana Project                                                                                                                                            | May '26 – Present                       |
| Youth Camp Counselor, Guru Nanak Sikh Heritage                                                                                                                                  | Aug '21 – Aug '25                       |
| President, Events Coordinator (Former), NJIT Sikhs                                                                                                                              | Dec '23 – May '26                       |
| President, Treasurer (Former), Co-Founder, NJIT MedicZero                                                                                                                       | Nov '23 – May '26                       |
| Co-Founder, Delta Epsilon Mu (Pre-medical Professional Fraternity)                                                                                                              | Feb '24 – May '25                       |
| Fellow, Honors Summer Research Institute Fellowship                                                                                                                             | Jun '24 – Jul '24,<br>Jun '25 – Jul '25 |
| EMT Senior Squad Member, Cadet Crew Chief (Former), Marlboro First Aid Squad                                                                                                    | Apr '22 – Jan '25                       |
| Fellow, No Patient Left Behind Fellowship                                                                                                                                       | Jun '24 – Aug '24                       |
| Young Scholar, American College of Cardiology                                                                                                                                   | May '23 – Apr '24                       |